

Garrett Brownlee
NSC 4400 01
Dr. Sanford
09/15/25

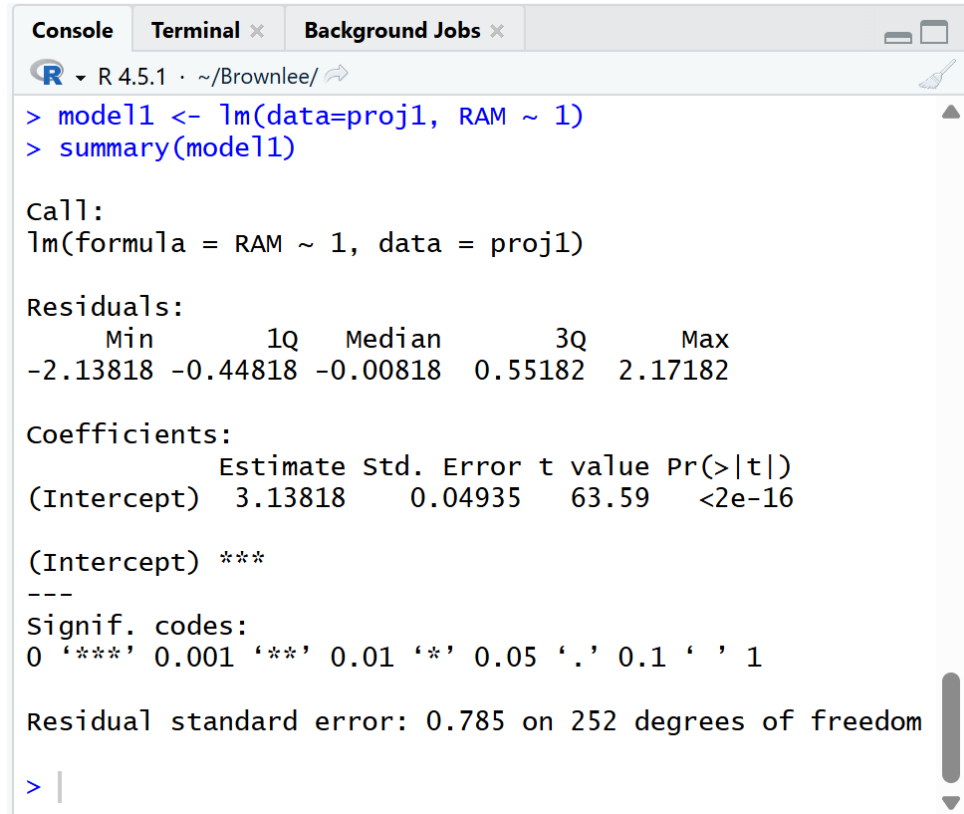
Step #4:

```
Console Terminal x Background Jobs x
R 4.5.1 · ~/Brownlee/
> Garrett <- c(2,5,2,3)
> Garrett - 3
[1] -1 2 -1 0
> Garrett^2
[1] 4 25 4 9
> sqrt(Garrett)
[1] 1.414214 2.236068 1.414214 1.732051
> sum(Garrett)
[1] 12
> length(Garrett)
[1] 4
> M <- sum(Garrett) / length(Garrett)
> SS <- sum((Garrett - M)^2)
> VAR <- SS / (length(Garrett) - 1)
> print(c(M, SS, VAR))
[1] 3 6 2
> sqrt(VAR)
[1] 1.414214
> mean(Garrett)
[1] 3
> sd(Garrett)
[1] 1.414214
>
```

Steps #5 and #6:

```
Console Terminal x Background Jobs x
R 4.5.1 · ~/Brownlee/
> proj1 <- rbind(proj1, c(318,1,2.9,NA))
> GBram <- proj1$RAM
> sum(GBram)
[1] 793.96
> length(GBram)
[1] 253
> Mram <- sum(GBram) / length(GBram)
> SSram <- sum((GBram - M)^2)
> VARram <- SS / (length(GBram) - 1)
> print(c(Mram, SSram, VARram))
[1] 3.13818182 160.10640000 0.02380952
> sqrt(VARram)
[1] 0.1543033
> mean(GBram)
[1] 3.138182
> sd(GBram)
[1] 0.7849668
>
```

Step #7:



```
Console Terminal x Background Jobs x
R 4.5.1 · ~/Brownlee/
> model1 <- lm(data=proj1, RAM ~ 1)
> summary(model1)

Call:
lm(formula = RAM ~ 1, data = proj1)

Residuals:
    Min       1Q   Median       3Q      Max
-2.13818 -0.44818 -0.00818  0.55182  2.17182

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.13818    0.04935   63.59  <2e-16

(Intercept) ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.785 on 252 degrees of freedom

> |
```

Step #8:

1. Add a written summary of the results for the mean and the standard deviation using the mean and standard deviation template from the lecture 1 notes posted on Canvas. When writing summaries, be sure to refer to the variables in your analysis using normal English words that people would understand even if they were not familiar with the dataset you are using. Never use the cryptic, abbreviated R variable names in your written summaries, and round all values to two decimal places.
 - **On a “Relationship Attribution Measure” scale that ranged from 1 to 6, the average score was 3.14. On average, people deviated from this score by .79 units.**
2. List the score your character received on the scale you picked in Step 1.
 - **2.9 (out of 6)**

- Using your character's score and the statistics from your screen shots, calculate your character's z-score and list your character's z-score in your document.

- **z-score** = $(2.9 - 3.138) / (.79) = -.30$

- Write a summary of your character's z-score using the z-score template from the lecture 1 notes posted on Canvas. Again, be sure to refer to the variables in your analysis using normal English words and not abbreviated code names.

- **My character's (Tom Cruize's) score on the "Relationship Attribution Measure" scale was .30 standard deviations below the mean. If the reported "Attribution" scores are normally distributed, his score would be between the 38th and 62nd percentile.**

- At the end of your Word document, paste a copy of the questionnaire (scanned or electronically completed) with your character's responses marked to show that the questionnaire was, indeed, completed. **SEE BELOW**

Garrett Brownlee (GBram)

The Relationship Attribution Measure (RAM)
Fincham, F. D., & Bradbury, T. N. (1992). Assessing attributions in marriage: The Relationship Attribution Measure. *Journal of Personality and Social Psychology*, 62, 457-468.

This scale will be most meaningful for people who are currently in a romantic relationship (or who were in a previous romantic relationship). It measures the extent to which you make negative relationship attributions. Negative attributions pertain to the explanations you make for causes of negative events in a relationship. Scoring high on this scale is associated with a risk for having high levels of conflict in a relationship. To complete this measure, think about a current (or previous) romantic relationship and imagine each of the following scenarios occurring in that relationship.

Imagine that your partner criticizes something you say. Rate how much you would agree with the following statements if your partner were to criticize something you say.

	Disagree strongly 1	Disagree 2	Disagree somewhat 3	Agree somewhat 4	Agree 5	Agree strongly 6
1. My partner is criticizing me due to something about him/her (e.g., the type of person he/she is, the mood he/she was in).			3			
2. The reason my partner criticized me is NOT likely to change.	1	2	3	4	5	6
3. The reason my partner criticized me is something that affects other areas of our relationship.	1	2	3	4	5	6
4. My partner criticized me on purpose rather than unintentionally.	1	2	3	4	5	6

Imagine that your partner begins to spend less time with you. How much would you agree with the following statements if your partner were to begin to spend less time with you?

	Disagree strongly 1	Disagree 2	Disagree somewhat 3	Agree somewhat 4	Agree 5	Agree strongly 6
5. My partner is spending less time with me due to something about him/her (e.g., the type of person he/she is, the mood he/she was in).			3			
6. The reason my partner is spending less time with me is NOT likely to change.	1	2	3	4	5	6
7. The reason my partner is spending less time with me is something that affects other areas of our relationship.	1	2	3	4	5	6
8. My partner is spending less time with me on purpose rather than unintentionally.	1	2	3	4	5	6

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Imagine that your partner does not pay attention to what you are saying. How much would you agree with the following statements if your partner were to not pay attention to what you are saying?

	Disagree strongly	Disagree	Disagree somewhat	Agree somewhat	Agree	Agree strongly
9. My partner is not paying attention due to something about him/her (e.g., the type of person he/she is, the mood he/she was in).	7	2	3	4	5	6
10. The reason my partner is not paying attention is NOT likely to change.	1	9	3	4	5	6
11. The reason my partner is not paying attention is something that affects other areas of our relationship.	1	2	3	4	5	6
12. My partner is not paying attention on purpose rather than unintentionally.	1	5	3	4	5	6

Imagine that your partner is cool and distant. How much would you agree with the following statements if your partner was cool and distant?

	Disagree strongly	Disagree	Disagree somewhat	Agree somewhat	Agree	Agree strongly
13. My partner is cool and distant due to something about him/her (e.g., the type of person he/she is, the mood he/she was in).	2	2	3	4	5	6
14. The reason my partner is cool and distant is NOT likely to change.	1	2	3	4	5	6
15. The reason my partner is cool and distant is something that affects other areas of our relationship.	1	2	3	4	5	6
16. My partner is cool and distant on purpose rather than unintentionally.	1	2	3	4	5	6

Sum = 46
Score = Sum / 16 = 2.88

To calculate your score on this scale, add your responses to all 16 items. **Then, divide that total by 16.** In other words, calculate your average response across all 16 items. This will produce a total negative attribution score on a scale that ranges from 1 to 6. A high score indicates greater levels of negative attributions.

Project Scoring Rubric:

Project 1 Scoring Table

1. **Part 4 screenshot.** This screenshot should match the example from the manual but show the student's name rather than "Keith."

Possible points	Deductions	Score
6	0	6

Criterion	Deduction
A. Section does not contain any content errors (missing steps or incorrect results).	3
B. Section contains no more than two content errors.	3
C. Screenshot size and resolution allow it to be read easily.	3
D. Screenshot cropped and does not show command lines with errors.	1

2. **Part 5 screenshot.** This screenshot should display all the steps and all the correct values using *two* methods to calculate a mean and standard deviation.

Possible points	Deductions	Score
6	0	6

Criterion	Deduction
A. Section does not contain any content errors (missing steps or incorrect results).	3
B. Section contains no more than two content errors.	3
C. Screenshot size and resolution allow it to be read easily.	3
D. Screenshot cropped and does not show command lines with errors.	1
E. A vector was created with a name ending with either "ram" or "pss."	3
F. Results show the number of cases to be 253 (and not 252, because if there are only 252 cases, it means that the step of adding an additional score was not completed).	3
G. Results show two methods for calculating a mean and standard deviation, and both methods produce the same value.	3

3. **Part 6 screenshot.** This screenshot should display the correct command and results for an analysis using a linear model to calculate a mean and standard deviation.

Possible points	Deductions	Score
6	0	6

Criterion	Deduction
A. Section does not contain any content errors (missing steps or incorrect results).	3
B. Section contains no more than two content errors.	3
C. Screenshot size and resolution allow it to be read easily.	3
D. Screenshot cropped and does not show command lines with errors.	1

E. The mean and standard deviation displayed on this page match the values from the previous screen shot.		3
4. Written summary of the mean and standard deviation.		
Possible points	Deductions	Score
20	0	20
Criterion		Deduction
A. The summary needs to name the scale being summarized. The name needs to be stated in plain language, and the summary should NOT contain any cryptic code names such as "FS 8."		5
B. The summary needs to state the correct range of scores. This range should NOT be described merely by stating something like, "it was a 7-point scale" (because this does not clarify the actual top and bottom values for a range of scores).		5
C. The summary needs to state the correct value for the mean.		5
D. The summary needs to make it clear that the student understands the meaning of a standard deviation. Specifically, this requires making a correct statement explaining that, on average, something deviates by some amount from something.		5
E. The summary needs to use the correct value for the standard deviation.		5
F. Values are rounded to two decimal places.		2
5. Written summary of a z-score.		
Possible points	Deductions	Score
20	0	20
Criterion		Deduction
A. The questionnaire score is placed in the correct location before the summary.		5
B. The questionnaire score appears some place in the project (without the questionnaire score, this section is not scorable).		20
C. The summary lists a correct value for the z-score.		5
D. The summary makes it clear that the student understands the meaning of a z-score. Specifically, this requires making a correct statement explaining how much something was above or below something. If the word "units" is used, there needs to be a clarification of what the units are (otherwise the statement will imply something incorrect).		5
E. The summary correctly specifies a percentile range where the score is presumed to fall.		5

F. The statement about the percentile range includes a phrase clarifying an assumption that was made to determine this range.	2
G. Values are rounded to two decimal places	2

6. Attachments.	Possible points	Deductions	Score
	3	0	3

Criterion	Deduction
A. A copy of a completed questionnaire is attached.	1
B. This grading table has been completed and is attached.	2